

15.1.1

Suppose X is a Banach space, $A \in \mathcal{C}(X)$, $T = I - A$. State the Riesz-Fredholm theorem and discuss the existence and uniqueness of solutions to the equation $Tx = y$.

Proof. (Riesz-Fredholm) Suppose X is a Banach space, $A \in \mathcal{C}(X)$, $T = I - A$. Then

- 1) $\sigma(T) = \sigma(T^*)$;
- 2) $\dim N(T) = \dim N(T^*) < \infty$;
- 3) $R(T) = N(T^*)^\perp$, $R(T^*) = {}^\perp N(T)$.

Uniqueness. The equation $Tx = y$ has at most one solution if and only if $N(T) = \{0\}$.

Existence. The equation $Tx = y$ has at least one solution if and only if $y \in N(T^*)^\perp$.

If $N(T) = \{0\}$, the equation $Tx = y$ has a unique solution for any $y \in X$. If $N(T) \neq \{0\}$, the equation has a solution if and only if $y \in N(T^*)^\perp$, and the solution set forms an affine space. $\square$

15.1.3

Suppose X is a infinite dimensional Banach space, $A \in \mathcal{C}(X)$. Then $\sigma(A) = \{0\} \cup \sigma_p(A)$, and for any $\delta > 0$, $\{\lambda \in \sigma_p(A) : |\lambda| > \delta\}$ is a finite set.

Proof. Since $\dim X > \infty$, $0 \in \sigma(A)$. Then $\sigma(A) = \{0\} \cup \sigma_p(A)$. Note that $\sigma(A)$ has at most 0 as its accumulation point, we have for any $\delta > 0$, $\{\lambda \in \sigma_p(A) : |\lambda| > \delta\}$ is a finite set. $\square$

15.2.2

Suppose A is a bounded self-adjoint operator on H , $m(A) = \inf_{\|x\|=1} \langle x, Ax \rangle$. Prove that

- 1) $\sigma(A) \subset [m(A), \|A\|]$;
- 2) if $A \in \mathcal{C}(H)$, $m(A) \neq 0$, then $m(A) \in \sigma_p(A)$.

Proof. 1) Let $M(A) = \sup_{\|x\|=1} \langle x, Ax \rangle$. Then $M(A) \leq \|A\|$. We may prove that $\sigma(A) \subset [m(A), M(A)]$. For any $\lambda < m(A)$, we have

$$\inf_{\|x\|=1} \langle (A - \lambda I)x, x \rangle = \inf_{\|x\|=1} \langle Ax, x \rangle - \lambda \geq m(A) - \lambda > 0.$$

This implies $R(A - \lambda I)$ is closed and $N(A - \lambda I) = \{0\}$. Since A is self-adjoint, we have

$$R(A - \lambda I)^\perp = N(A^* - \bar{\lambda} I^*) = N(A - \lambda I) = \{0\}.$$

Then $A - \lambda I$ has a bounded inverse, implies $\lambda \notin \sigma(A)$. For $\lambda > M(A)$, the proof is similar. Note that $\sigma(A) \subset \mathbb{R}$, we have $\sigma(A) \subset [m(A), M(A)] \subset [m(A), \|A\|]$.

2) Since $A \in \mathcal{C}(H)$, by Hilbert-Schmidt Thm, there exists at most countable non-zero real numbers $\{\lambda_i\}$ and an orthonormal basis $\{e_i\}$ such that for any $x \in H$,

$$Ax = \sum_{i \in I} \lambda_i \langle x, e_i \rangle e_i.$$

Then we have

$$\inf_{\|x\|=1} \langle Ax, x \rangle = \inf_{\|x\|=1} \sum_{i \in I} \lambda_i \langle x, e_i \rangle^2 \geq \inf_{i \in I} \lambda_i,$$

and

$$\lambda_i = \langle Ae_i, e_i \rangle \geq \inf_{\|x\|=1} \langle Ax, x \rangle = m(A).$$

Hence $m(A) = \inf_{i \in I} \lambda_i$. Note that $m(A) \neq 0$ and $\sigma(A)$ has at most 0 as its accumulation point, we have $m(A) \in \sigma_p(A)$. $\square$

15.2.4

Suppose A is a non-negative compact self-adjoint operator on H . The eigenvalues listed with multiplicities as $\lambda_1 \geq \lambda_2 \geq \cdots \geq 0$. Prove that

$$\lambda_n = \inf_{E_{n-1} \subset H} \sup_{x \in E_{n-1}^\perp \setminus \{0\}} \frac{\langle Ax, x \rangle}{\|x\|^2},$$

where E_{n-1} is the subspace of H satisfying $\dim E_{n-1} = n - 1$.

Proof. For any $E_{n-1} \subset H$, there exists $x_0 \in \text{span} e_1, e_2, \dots, e_n$, $x_0 \neq 0$ such that $x_0 \in E_{n-1}^\perp$. Then

$$\sup_{x \in E_{n-1}^\perp \setminus \{0\}} \frac{\langle Ax, x \rangle}{\|x\|^2} \geq \frac{\langle Ax_0, x_0 \rangle}{\|x_0\|^2} \geq \lambda_n.$$

Hence

$$\inf_{E_{n-1} \subset H} \sup_{x \in E_{n-1}^\perp \setminus \{0\}} \frac{\langle Ax, x \rangle}{\|x\|^2} \geq \lambda_n.$$

Let $E_{n-1} = \text{span} e_1, e_2, \dots, e_{n-1}$, then

$$\lambda_n \geq \sup_{x \in E_{n-1}^\perp \setminus \{0\}} \frac{\langle Ax, x \rangle}{\|x\|^2}.$$

This implies

$$\lambda_n = \inf_{E_{n-1} \subset H} \sup_{x \in E_{n-1}^\perp \setminus \{0\}} \frac{\langle Ax, x \rangle}{\|x\|^2}.$$

□